

Heisenberg-scaling sparse Hamiltonian learning with a fixed minimum query duration

Yuxuan Zhang

Institute of Physics, École Polytechnique Fédérale de Lausanne (EPFL),
CH-1015 Lausanne, Switzerland

Department of Physics, Princeton University,
Princeton, New Jersey 08544, USA

1 October 2026 — Version 1.0

Abstract

We give an algorithm for learning an unknown sparse Pauli Hamiltonian when every query to its evolution must last at least a prescribed constant $T > 0$. For a traceless n -qubit Hamiltonian of operator norm at most one and at most m nonzero Pauli coefficients, the algorithm achieves maximum coefficient error ε with probability at least $2/3$. Its total unknown evolution time and query count are $\tilde{O}_T(\text{poly}(m)/\varepsilon)$, and its known circuit and classical-processing costs are polynomial in n, m, ε^{-1} . No locality assumption or advance knowledge of the Pauli support is required. A coarse learner first discovers relevant labels from long-time Choi states. A two-time filter then inverts the linear response of the residual Hamiltonian: the two response functions have no common spectral zero on the required interval. Implementing this filter with a bounded sum of known unitaries allows repeated residual queries to improve precision at Heisenberg scaling. All unknown queries are ordinary forward evolutions; controlled unknown evolution and inverse access are unnecessary. The constants and polynomial degrees may depend on fixed T .

Draft status. The underlying candidate and resource supplement passed internal model-assisted reviews. This manuscript presents their argument in expanded form. Literature priority and external expert confirmation remain unestablished.

1 The question and the result

Sparse Hamiltonian learning asks for the strengths and labels of a small number of unknown interactions. Short evolution intervals make those interactions directly visible in the linear term of the propagator. If each query has a positive minimum duration, that approximation is unavailable: different spectral components can accumulate phases that cancel the response to an interaction. The issue is whether this loss of a short-time probe necessarily costs either exponential dependence on sparsity or more than inverse-linear dependence on precision.

We resolve the fixed-minimum-duration, polynomial-sparsity question posed by Shin, Lee, and Oh [2], in the oracle model specified below. The construction uses a coarse estimate to compensate the known part of the Hamiltonian. Two allowed evolution durations then provide a uniformly invertible response to the remaining error. The inverse acts on a Choi state through known controls, and its normalization is independent of the Hilbert space dimension and the target precision.

Let $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$, with Hermitian Pauli representatives, and write

$$H = \sum_{P \in \mathcal{P}_n \setminus \{I\}} a_P P, \quad |\text{supp}(a)| \leq m, \quad \|H\|_{\text{op}} \leq 1. \quad (1)$$

The coefficients are real, and their support is unknown. The learner knows an upper bound $m \geq 1$. An oracle supplies $U(t) = e^{-iHt}$ for any chosen $t \geq T$, where $T > 0$ is fixed independently of n, m, ε . Known quantum controls and ancillas may be used between oracle calls. Total unknown evolution time means $\sum_{\ell} t_{\ell}$, summed over every call, including calls in failed filtering experiments.

Theorem 1. *For every fixed $T > 0$, there is an algorithm which, for every Hamiltonian satisfying (1) and every $0 < \varepsilon < 1$, returns a list of at most m Pauli labels and real coefficients such that*

$$\Pr\left\{\max_{P \neq I} |\hat{a}_P - a_P| \leq \varepsilon\right\} \geq \frac{5}{6}.$$

Unlisted coefficients are interpreted as zero. The number of unknown queries and their total evolution time are

$$\tilde{O}_T\left(\frac{\text{poly}(m)}{\varepsilon}\right).$$

Known circuit and classical-processing costs are polynomial in n, m, ε^{-1} , on every measurement history. Each unknown query is an ordinary forward evolution of duration at least T . The construction uses neither a controlled unknown oracle nor an unknown inverse.

The notation $\tilde{O}_T$ suppresses logarithmic factors in m and ε^{-1} , and constants depending on fixed T . The degree of the initialization polynomial can also depend on T . In particular, when $m \leq n^c$ for fixed c , all costs other than the displayed evolution resource remain polynomial in n, ε^{-1} .

The timing model matters. Two allowed durations may differ by a small amount even though both exceed T . The theorem does not assume a fixed timing grid or a lower bound on the separation of programmable durations. It also charges known gate work separately from unknown evolution time; the known gate count need not be linear in ε^{-1} .

1.1 Relation to previous algorithms

Hu et al. [1] obtain Heisenberg scaling for sparse Hamiltonians with unknown interaction structure. Their Hamiltonian-reshaping procedure interleaves short unknown evolutions with Pauli controls; see their Eq. (9) and the error estimate following Eq. (11). Shin, Lee, and Oh [2] treat a minimum duration explicitly. Their Theorem 1 has an exponential 4^m sparsity factor and is efficient for logarithmic sparsity at fixed T . Their Theorem 2 trades a minimum duration $T = \Theta(m^{-1/K})$ against a sparsity factor m^{K+2} in the Heisenberg term. The subsequent discussion asks for polynomial sparsity dependence at arbitrary constant T . We retain the iterative estimation-and-compensation architecture. The additional ingredient here is a bounded two-time inverse that avoids enumeration of the algebra generated by the unknown Pauli support.

Two other long-time access models address different identification tasks. Cedillo Vayson de Pradenne, Cotler, and Huang [3] recover a Hamiltonian direction under generic local identifiability assumptions; that guarantee does not give worst-case absolute coefficients for arbitrary nonlocal sparse support. Zhou and Gong [4] give a control-free, ancilla-free in-situ learner with inverse-quadratic precision dependence in total evolution time. Our algorithm uses ancillas and known interleaved controls.

Shin and Tong [5] subsequently develop eigenphase engineering. Their Theorem 7, Eq. (35), uses $\tau = \Theta([\Lambda(1 + \log(\Lambda/\varepsilon))]^{-1})$, which decreases with the requested error. Their sparse-learning

Theorem 9 also assumes fixed locality, while Theorem 12 estimates a specified coefficient using the same energy-gap interface. Those stated results do not supply the arbitrary-support, fixed- T guarantee of Theorem 1. These comparisons concern the cited versions and do not constitute exhaustive historical priority clearance.

1.2 Notation and organization

Put $d = 2^n$ and

$$|\Phi_d\rangle = \frac{1}{\sqrt{d}} \sum_{x=0}^{d-1} |x\rangle|x\rangle, \quad |X\rangle = (X \otimes I)|\Phi_d\rangle.$$

Thus $\langle X | Y \rangle = d^{-1} \text{Tr}(X^\dagger Y)$. We write $\|X\|_{2,d} = (d^{-1} \text{Tr} X^\dagger X)^{1/2}$, so that $\|X\|_{2,d} \leq \|X\|_{\text{op}}$. The vectors $|P\rangle$, $P \in \mathcal{P}_n$, form the Pauli–Bell basis; measuring it requires only known Clifford gates and computational-basis measurements. For any operator X , its Pauli coefficient is $X_P = d^{-1} \text{Tr}(PX)$.

Fix an integer $\tau \geq \max\{1, T\}$. All constants constructed below depend only on τ . Section 2 gives a coarse estimate at polynomial cost. Sections 3 and 4 construct and measure the filtered residual. Section 5 combines the estimates and accounts for all resources. Appendix A gives simultaneous explicit choices of the constants.

2 A coarse estimate from allowed durations

The initial estimate need only have inverse-polynomial operator error in m ; it need not attain the final precision. This permits a polynomially costly extrapolation from allowed durations to the derivative at zero.

Lemma 2. *For $0 < z \leq 1$ and $0 < p < 1/2$, one can construct an at-most- m -term traceless estimate H_0 such that, except with probability at most $2p$,*

$$\max_{P \neq I} |(H_0 - H)_P| \leq 2z, \quad \|H_0 - H\|_{\text{op}} \leq 4mz.$$

The query, evolution-time, known-circuit and classical costs are polynomial in $n, z^{-1}, \log(m/p)$ for fixed τ . Every oracle duration lies in $[\tau, \tau + 1]$.

Proof. Define $f_P(t) = d^{-1} \text{Tr}(Pe^{-iHt})$. Every real-time derivative of f_P has modulus at most one, because $\|H\|_{\text{op}} \leq 1$, and $f'_P(0) = -ia_P$ for $P \neq I$. Let $b_0 = \tau + 1$, take nodes $t_j = \tau + j/L$ for $0 \leq j \leq L$, and let $w_j = \ell'_j(0)$ be the derivative weights of the Lagrange cardinal polynomials at those nodes. The denominator of ℓ_j has modulus $j!(L-j)!/L^L$. Differentiating its numerator therefore gives

$$W := \sum_{j=0}^L |w_j| \leq \frac{Lb_0^{L-1}L^L2^L}{L!} \leq \frac{L}{b_0} (2eb_0)^L. \quad (2)$$

Interpolation differentiates the degree- L Taylor polynomial exactly. The integral remainder along the real axis consequently yields

$$|f'_P(0) - \sum_{j=0}^L w_j f_P(t_j)| \leq \frac{Wb_0^{L+1}}{(L+1)!} \leq \left(\frac{2e^2b_0^2}{L} \right)^L. \quad (3)$$

Choose an integer

$$L \geq \max\{4e^2b_0^2, \log_2(4/z)\}.$$

The last expression is at most $z/4$. If $|a_P| \geq z$, some node satisfies $|f_P(t_j)| \geq 3z/(4W)$. Choose a node uniformly, prepare $|U(t_j)\rangle$, and measure in the Pauli–Bell basis. The probability of sampling this label is at least

$$\frac{9z^2}{16(L+1)W^2}.$$

After $O((L+1)W^2z^{-2}\log(2m/p))$ repetitions, a union bound over at most m heavy labels shows that all of them have been seen with probability at least $1-p$. Keep the distinct nonidentity labels obtained. Their number L_c is at most the number of shots, without enumerating the 4^n possible Pauli strings. Since $L = O_\tau(1 + \log(1/z))$, Eq. (2) makes this cost polynomial in z^{-1} for fixed τ .

To estimate a sampled coefficient, prepare two independent Choi states $x = |U(\tau)\rangle$ and $y = |U(\tau + \delta)\rangle$. Let $\mathbf{S}$ swap the entire two Choi registers and let $\mathbf{P}$ act by P on the output half of the first register. The known unitary $\mathbf{PS}$ obeys

$$\begin{aligned} \langle x, y | \mathbf{PS} | x, y \rangle &= \langle U(\tau) | PU(\tau + \delta) \rangle \langle U(\tau + \delta) | U(\tau) \rangle \\ &= f_P(\delta) \overline{f_I(\delta)}. \end{aligned} \quad (4)$$

The second identity uses cyclicity of the trace and the common Hamiltonian of the two evolutions. A Hadamard test estimates its imaginary part by controlling only $\mathbf{PS}$.

Tracelessness gives

$$|f_P(\delta) + i\delta a_P| \leq \frac{\delta^2}{2}, \quad |f_I(\delta) - 1| \leq \frac{\delta^2}{2}.$$

Using $|f_P(\delta)| \leq 1$, it follows that minus the imaginary part of (4), divided by δ , has bias at most δ as an estimate of a_P . Set $\delta = z/2$. Allocate statistical coefficient error $z/4$ and implementation error $z/4$, in addition to the bias $z/2$. Hoeffding’s inequality then gives simultaneous coefficient error at most z , except with probability p , using $O(z^{-4}\log(2\max\{1, L_c\}/p))$ fresh shots for each label. Both oracle durations are at least τ ; a small δ is a difference of durations, not an unknown evolution of duration δ .

Retain the m largest estimated magnitudes, or all labels if fewer were sampled. A retained coefficient has error at most z . An omitted true label absent from the list has magnitude below z . If a true label in the list is omitted, one of the retained labels must be spurious: there are at most m true labels in total. Its estimate has magnitude at most z , and hence the omitted true coefficient has magnitude at most $2z$. The union of the true and retained supports has size at most $2m$. The coefficient and operator bounds follow.

All list operations use Pauli strings of length n . The Bell discovery measurement uses exact Clifford gates. Known gates in the two-copy tests can be compiled so that each estimated expectation changes by at most $z^2/8$. Division by $\delta = z/2$ converts this to at most $z/4$, as allocated above, with polynomial known-gate work. $\square$

3 Inverting the response at two allowed times

Suppose now that H_0 is known, traceless, and at most m -sparse, with

$$\|H_0\|_{\text{op}} \leq 2, \quad \Delta = H - H_0, \quad \|\Delta\|_{\text{op}} \leq \eta.$$

On the operator Hilbert space, set $A = \text{ad}_{H_0}$, so $AX = H_0X - XH_0$. This is self-adjoint and has spectrum in $[-4, 4]$. Its unitary group is conjugation by known evolution:

$$e^{ivA}X = e^{ivH_0}Xe^{-ivH_0}.$$

Choose $h = 1/4$, $t_1 = \tau$, $t_2 = \tau + h$, and $b = t_2$. Define

$$F_j(x) = \int_0^{t_j} e^{ixs} ds, \quad G(x) = \frac{|F_1(x)|^2 + |F_2(x)|^2}{2}, \quad R_j(x) = \frac{\overline{F_j(x)}}{G(x)}. \quad (5)$$

Lemma 3. *On $[-6, 6]$,*

$$G(x) \geq \frac{h^2}{\pi^2}, \quad \frac{R_1(x)F_1(x) + R_2(x)F_2(x)}{2} = 1.$$

In particular R_j is smooth on this interval and bounded by a constant depending only on τ .

Proof. For $x \neq 0$, put $u = xt_1/2$ and $v = xh/2$. Then

$$G(x) = \frac{2}{x^2} (\sin^2 u + \sin^2(u + v)) \geq \frac{2(1 - |\cos v|)}{x^2}.$$

Here $|v| \leq 3/4$, so $\cos v > 0$. The inequality $\sin y \geq 2y/\pi$ on $[0, \pi/2]$ gives $1 - \cos v \geq 2v^2/\pi^2$, proving the lower bound. At zero it follows by continuity; directly, $G(0) = (t_1^2 + t_2^2)/2$. The inverse identity follows from the definition. Since $|F_j| \leq b$, the asserted boundedness follows. $\square$

Although an individual F_j can vanish, their squared magnitudes have a uniform positive sum. The inverse can therefore be implemented without resolving the eigenvalues of H_0 .

Lemma 4. *There is a rational constant $S > 1$, depending only on τ , with the following property. For $0 < \rho \leq 1$, one can compute real rational coefficients defining*

$$Q_j(x) = \sum_{|k| \leq D} q_{jk} e^{i\pi k x/8}, \quad D = O_\tau(\rho^{-1}),$$

such that

$$\sup_{|x| \leq 4} |Q_j(x) - R_j(x)| \leq \rho, \quad Q_j(0) = \kappa_j := R_j(0) = \frac{2t_j}{t_1^2 + t_2^2}, \quad \sum_k |q_{jk}| < S.$$

A known quantum circuit has successful-outcome Kraus operator $K_j = Q_j(A)/S$ on the Choi space. In particular,

$$K_j|I\rangle = k_j|I\rangle, \quad k_j = \kappa_j/S > 0, \quad \|K_j\|_{\text{op}} \leq 1.$$

The coefficients can be computed in polynomial time in ρ^{-1} for fixed τ . If the Pauli coefficient sum of H_0 is at most $2m$, the circuit has polynomial known-gate cost at inverse-polynomial implementation accuracy.

Proof. Take an even C^2 cutoff χ , equal to one on $[-4, 4]$ and zero outside $[-6, 6]$. On $4 < |x| < 6$, set $\chi(x) = 1 - 10u^3 + 15u^4 - 6u^5$, where $u = (|x| - 4)/2$. Extend χR_j periodically with period 16. Its Fourier coefficients c_{jk} are real because $R_j(-x) = \overline{R_j(x)}$. Two integrations by parts give

$$|c_{jk}| \leq \frac{4\|(\chi R_j)''\|_{L^1}}{\pi^2 k^2} \quad (k \neq 0). \quad (6)$$

The derivative norm is bounded solely in terms of τ ; explicit bounds are in Appendix A. Thus the Fourier coefficients have a bounded absolute sum and a tail of order $O_\tau(D^{-1})$.

Truncate at $D = O_\tau(\rho^{-1})$ and approximate the coefficients by rationals with total error at most $\rho/8$. Choose the tail bound at most $\rho/4$. The resulting polynomial differs from R_j by at most $3\rho/8$

on $[-4, 4]$. Adjust its constant coefficient to make its value at zero exactly κ_j . This adds at most $3\rho/8$ to the error and preserves a uniform bound on the absolute coefficient sum. The adjustment is rational. Standard certified quadrature on a mesh of order $O_\tau(\rho/D^2)$ suffices for the stated total coefficient error; the coefficient integrands have derivative bounded by $O_\tau(1 + D)$.

Choose rational S larger than these absolute sums and append two opposite identity coefficients to bring the sum exactly to S . Preparing amplitudes $\sqrt{|q_{jk}|/S}$, selecting the signed unitaries $e^{i\pi k A/8}$, and unpreparing the coefficient register gives successful Kraus operator $Q_j(A)/S$. The selected known unitary on the Choi space is

$$e^{i\pi k H_0/8} \otimes e^{-i\pi k H_0^T/8}.$$

Because $A|I\rangle = 0$, the action on $|I\rangle$ is exact before finite gate compilation. Contractivity follows directly from the absolute coefficient sum.

For completeness, a known Pauli Hamiltonian with coefficient absolute sum at most $2m$ can be simulated for duration v to operator error γ by a first-order product formula with $O(1 + m^2 v^2/\gamma)$ slices, each containing at most m Pauli rotations. The bound follows by the two-term Duhamel commutator estimate and telescoping products of unitaries. Each Pauli rotation costs polynomially many elementary gates in n and its required precision. Here $|v| \leq \pi D/8$. Using a circuit V and its complex conjugate as the pair $V \otimes \bar{V}$ preserves $|I\rangle$. Coefficient-register preparation and selection also have polynomial cost in D, n, m, γ^{-1} . Finite-precision errors will be budgeted for the complete experiment below. $\square$

4 Amplification and measurement of the residual

Form the compensated evolution

$$W_j = e^{iH_0 t_j} e^{-iH t_j}.$$

One application uses exactly one allowed unknown query. The compensator is a known unitary, even though it has the opposite sign in its exponent. Repeating this experiment accumulates the small residual rather than requiring a shorter unknown evolution.

Lemma 5. *For any positive integer r , writing $q = r\eta$ and $B = b^2/2$,*

$$W_j^r = I - irF_j(A)\Delta + E_j, \quad \|E_j\|_{\text{op}} \leq Bq^2.$$

Proof. The interaction-picture propagator for one block has generator $e^{iH_0 s} \Delta e^{-iH_0 s}$ on $0 \leq s \leq t_j$. Represent the r identical blocks as a time-ordered propagator with this generator repeated on r consecutive intervals. Its first integral is $rF_j(A)\Delta$ and the integral of its operator norm is at most $rt_j\eta$. In the second-order integral remainder, leave the final propagator unexpanded. It is unitary, so the remainder norm is at most $(rt_j\eta)^2/2 \leq Bq^2$. $\square$

Apply the filter of Lemma 4 to $|W_j^r\rangle$. Let $y_j = K_j|W_j^r\rangle$ denote the unnormalized successful vector, and put

$$d_{jP} = (R_j(A)F_j(A)\Delta)_P.$$

The multiplier $R_j F_j = |F_j|^2/G$ is real and even. It preserves Hermiticity, so d_{jP} is real, and

$$\frac{d_{1P} + d_{2P}}{2} = \Delta_P, \quad |d_{jP}| \leq Mb\eta,$$

where M is any common bound on $|R_j|$ on $[-6, 6]$. Lemma 5 and the approximation error of Q_j give

$$y_{jI} = k_j + e_{jI}, \quad y_{jP} = -\frac{ir}{S}d_{jP} + e_{jP} \quad (P \neq I), \quad (7)$$

$$\|e_j\|_2 \leq E := Bq^2 + \frac{\rho bq}{S}, \quad |e_{jI}| \leq k_j Bq^2. \quad (8)$$

The sharper identity-component bound uses tracelessness of the linear term and $Q_j(0) = \kappa_j$. It has no additional filter-approximation term.

4.1 Discovering residual labels

Choose j uniformly from $\{1, 2\}$, prepare the filtered state, and measure the Pauli–Bell basis on filter success. On failure, return no label. Jensen’s inequality applied to the two unnormalized amplitudes in (7) gives

$$\Pr\{P\} = \frac{1}{2} \sum_{j=1}^2 |y_{jP}|^2 \geq \left(\frac{r|\Delta_P|}{S} - E \right)^2 \quad (9)$$

whenever the parenthesis is positive. This is an unconditional probability; postselection success has already been charged.

4.2 Estimating signed coefficients

For $P \neq I$, measure on success the Hermitian observable

$$O_P = -i|I\rangle\langle P| + i|P\rangle\langle I|.$$

Its outcomes are $+1, -1, 0$. Assign score zero to filter failure. The unconditional mean score is $2\text{Im}(\bar{y}_{jI} y_{jP})$. For the observed score Z , return

$$X = -\frac{S}{2rk_j} Z.$$

The ideal mean is d_{jP} ; randomizing j therefore targets Δ_P . Expanding the product in (7) yields

$$\begin{aligned} |\mathbb{E}X - \Delta_P| &\leq \frac{SE}{r} + Bq^2 M b \eta + \frac{SBq^2 E}{r} \\ &\leq C_0 \eta (q + \rho) \quad (q, \rho \leq 1), \end{aligned} \quad (10)$$

$$|X| \leq C_1/r, \quad C_1 = \frac{S}{2 \min_j k_j}. \quad (11)$$

One valid C_0 is specified in Appendix A. To implement O_P , first transform from the Bell basis to the computational basis, distinguish the two known strings corresponding to I and P , and rotate their two-dimensional span. This requires polynomially many known gates in n .

Proposition 6. *There are computable constants $K \geq 1$, $c > 0$, depending only on τ , for which the following holds. Set*

$$a = \frac{1}{Km}, \quad 0 < \eta \leq a/2, \quad r = \left\lfloor \frac{a}{\eta} \right\rfloor, \quad \rho = \frac{c}{m}, \quad \zeta = \frac{\eta}{16m}.$$

Using $\tilde{O}_\tau(m^8)$ experiments, with logarithmic dependence on the inverse failure probability, one can discover every residual label of magnitude at least ζ and estimate all discovered labels, together with the labels of H_0 , to error at most ζ . The discovery and estimation failure probabilities are each at most p . Every experiment uses exactly r unknown calls.

Proof. The stated choices imply $a/2 \leq q = r\eta \leq a$. Constants can be chosen simultaneously so that

$$E \leq \frac{q}{64mS}, \quad C_0(q + \rho) \leq \frac{1}{64m}. \quad (12)$$

Then for $|\Delta_P| \geq \zeta$, Eq. (9) is at least

$$\left(\frac{3q}{64mS}\right)^2 \geq \frac{c_2}{m^4}, \quad c_2 = \left(\frac{3}{128KS}\right)^2.$$

The residual has at most $2m$ nonzero labels. Therefore $O_\tau(m^4 \log(4m/p))$ discovery experiments suffice. Adjoin all current labels of H_0 . The candidate count is $L_c = O_\tau(m^4 \log(4m/p))$.

The bias in (10) is at most $\zeta/4$. Moreover $r\zeta \geq 1/(32Km^2)$. Using fresh data, Hoeffding's inequality and (11) give statistical error at most $\zeta/2$ on all L_c labels with $O_\tau(L_c m^4 \log(4 \max\{1, L_c\}/p))$ experiments. This holds conditional on any preceding discovery history.

It remains to justify finite gate compilation. Choose $c_3 > 0$ as in Appendix A and implement each complete experiment within trace distance $\nu = c_3/m^4$ of its ideal output, including its success or failure flag. A label probability changes by at most ν , so the discovery probability remains at least $c_2/(2m^4)$. A score mean changes by at most 2ν , and hence a coefficient estimate changes by at most $2C_1\nu/r \leq \zeta/4$. The total estimation error is therefore at most ζ .

Compile each compensator to operator error $O(\nu/r)$, and allocate a total $O(\nu)$ error to the remaining preparation, filter and measurement gates. Telescoping the r blocks gives the required trace-distance bound. The simulation in Lemma 4, finite precision coefficient preparation and known gate synthesis attain these accuracies with polynomial cost in n, m, r . There is no additional unknown query for implementing the filter. $\square$

5 Bootstrap, success probability, and total cost

Let

$$\eta_0 = \frac{1}{2Km}, \quad z = \frac{\eta_0}{4m}.$$

Run Lemma 2 at this value of z . On its success event, the resulting estimate has operator error at most η_0 , and therefore operator norm at most two. If $\varepsilon \geq \eta_0$, return that estimate. Otherwise define $\eta_s = \eta_0/2^s$, and let J be the smallest integer such that $\eta_J \leq \varepsilon$.

At each stage $s < J$, apply Proposition 6 to the current estimate with $\eta = \eta_s$. Add the estimated residual coefficients to the current coefficients, and retain the m largest magnitudes. Every listed provisional coefficient has error at most $\zeta = \eta_s/(16m)$. Outside the list, H_0 is zero and every true coefficient has magnitude below ζ , since all heavy residual labels were discovered. The truncation argument of Lemma 2 therefore gives

$$\|H - H_{s+1}\|_{\text{op}} \leq 4m\zeta = \eta_s/4 \leq \eta_{s+1}.$$

This proves the induction. At termination, $|(H_J - H)_P| \leq \|H_J - H\|_{\text{op}} \leq \varepsilon$ for every P .

Choose $p = 1/[12(J+1)]$, with $J = 0$ when no refinement is needed. There are two failure events in initialization and two at each refinement stage. Applying the bounds conditional on all earlier successes, a union bound gives total failure probability at most $2(J+1)p = 1/6$.

The number of experiments at each refinement stage is $\tilde{O}_\tau(m^8)$. Each experiment uses $r_s \leq a/\eta_s$ queries, whose durations belong to $\{\tau, \tau + 1/4\}$. For $J \geq 1$, $\eta_{J-1} > \varepsilon$, and hence

$$\sum_{s=0}^{J-1} r_s \leq a \sum_{s=0}^{J-1} \eta_s^{-1} < \frac{2a}{\varepsilon}.$$

Thus the refinement query count and unknown evolution time are conservatively bounded by $\tilde{O}_\tau(m^8/\varepsilon)$. This exponent is not optimized.

Initialization has $z^{-1} = 8Km^2$ and $L = O_\tau(1 + \log m)$. Equation (2) makes its entire query and evolution cost a polynomial in m , up to the failure logarithms. If P_τ denotes such a polynomial, the more informative total bound is

$$\tilde{O}_\tau\left(P_\tau(m) + \frac{m^8}{\varepsilon}\right). \quad (13)$$

In particular, the initialization is not being asserted to have an m^8 bound uniform in τ . Equation (13) implies the resource statement of Theorem 1.

We finally bound computation on unsuccessful measurement histories. Abort and output any fixed list if a retained coefficient exceeds two in magnitude. This never occurs on the success events used above: every true coefficient has modulus at most one, and the proved errors are smaller than one. On every continued history the estimate has at most m terms and Pauli coefficient absolute sum at most $2m$. Even when its operator norm is larger than two, the same finite filter circuit exists and its cost is bounded by that coefficient sum. The spectral approximation need only hold on good histories; the circuit does not test the operator norm or diagonalize the estimate.

The number of stages, shots and stored labels can be capped in advance by the displayed bounds. The constants S, κ_j are rational, and the score factors are rational, so all coefficient updates can be kept as exact rational sums with polynomial bit length. Known simulations use polynomially many slices and gates at the required inverse-polynomial accuracies in n, m, ε^{-1} . This establishes polynomial classical and circuit costs on all histories and completes the proof of Theorem 1.

6 What the construction establishes

The obstruction from a single long-time response is removed by a pair of response functions with a common inverse after averaging. The inverse has a bounded Fourier coefficient sum, and its action on the identity component is fixed exactly before compilation. These two properties make both label sampling and signed coefficient estimation possible at a precision-independent polynomial number of experiments per refinement stage. The inverse-linear precision dependence comes from geometrically increasing the number of compensated queries in each experiment.

The theorem concerns a programmable ideal oracle with known controls. It does not establish robustness to unknown implementation noise, a timing grid, or a restriction to local controls. Its conservative constants and initialization degree can be large at large fixed T . Reducing these costs is a separate problem from establishing polynomial dependence for each fixed minimum duration.

A Explicit constants and numerical precision

This appendix makes the choices in the proof simultaneous and effective. Let $b = \tau + 1/4$, $B = b^2/2$, and use the rational lower bound $g_0 = 1/160 < h^2/\pi^2$. Set

$$\begin{aligned} M &= b/g_0, \\ D_1 &= \frac{b^2}{2g_0} + \frac{b^4}{g_0^2}, \\ D_{2,\text{raw}} &= \frac{b^3}{3g_0} + \frac{13b^5}{6g_0^2} + \frac{2b^7}{g_0^3}, \\ D_2 &= 12(D_{2,\text{raw}} + 2D_1 + 4M). \end{aligned}$$

Indeed, $|F_j^{(\ell)}| \leq b^{\ell+1}/(\ell+1)$, $|G'| \leq b^3$, and $|G''| \leq 7b^4/6$. Differentiating $R_j = \overline{F_j}/G$ gives $|R'_j| \leq D_1$ and $|R''_j| \leq D_{2,\text{raw}}$. The cutoff satisfies $|\chi'| < 1$, $|\chi''| < 4$ and has support of length 12, so $\|(\chi R_j)''\|_{L^1} \leq D_2$.

The Fourier absolute sum is at most $M + 4D_2/3$. Choose an integer $S > M + 2D_2 + 2$. Taking an integer $D \geq 32D_2/\rho$ gives

$$\sum_{|k|>D} |c_{jk}| \leq \frac{8D_2}{\pi^2 D} \leq \frac{\rho}{4}.$$

Rational approximation with total coefficient error $\rho/8$ and the exact adjustment at zero leave uniform error at most $3\rho/4 < \rho$ and absolute coefficient sum below S . The padding with two opposite identity coefficients can consequently be done with nonnegative rational magnitudes.

For the estimator, choose

$$C_0 = SB(1+B) + BMb + b(1+B), \quad k_{\min} = \min_j \frac{\kappa_j}{S}, \quad C_1 = \frac{S}{2k_{\min}}.$$

The first line of (10) is bounded by

$$\eta[(SBq + b\rho)(1 + Bq^2) + BMbq^2],$$

which gives its second line for $q, \rho \leq 1$. Take an integer

$$K \geq \max\{1, 128SB, 128C_0\}$$

and a positive rational

$$c \leq \min\left\{1, \frac{1}{128b}, \frac{1}{128C_0}\right\}.$$

With the parameters of Proposition 6,

$$\frac{E}{q} \leq \frac{1}{m} \left(\frac{B}{K} + \frac{bc}{S} \right) \leq \frac{1}{64mS}, \quad C_0(q + \rho) \leq \frac{1}{64m}.$$

Finally set $c_2 = (3/(128KS))^2$ and choose rational

$$0 < c_3 \leq \min\left\{1, \frac{c_2}{2}, \frac{1}{256KC_1}\right\}.$$

For $\nu = c_3/m^4$, the compilation-induced estimation error obeys

$$\frac{2C_1\nu/r}{\zeta} \leq \frac{64KC_1c_3}{m^2} \leq \frac{1}{4}.$$

All constants are computable rational functions or integer bounds in the chosen fixed integer τ . Elementary-function evaluation for quadrature and rotation synthesis needs only polynomial bit precision. No spectrum of the unknown Hamiltonian or of its estimate is an input to the construction.

Human and AI contributions

The author set the research directions, designed the harness that coordinated model tasks, tools, memory, and checking, supplied physical and mathematical intuitions and proof skeletons, and guided the development and revision of the research ideas. AI models developed detailed candidate

arguments and derivations, searched the literature, wrote and ran computational checks, and drafted and revised the manuscript under this supervision.

For this result, the coordinating assistant proposed the paired-Choi and filtered-residual route, supplied explicit resource constants, and checked the source interfaces. The completed candidate and its resource supplement were reviewed in fresh contexts within the same model family. Those checks are internal model-assisted review, rather than independent human expert confirmation. The analytic argument, not the finite numerical controls, supports the universal statement. The source comparison uses the versions listed below and was revisited on 1 October 2026.

References

- [1] H.-Y. Hu, M. Ma, W. Gong, Q. Ye, Y. Tong, S. T. Flammia, and S. F. Yelin, *Ansatz-free Hamiltonian learning with Heisenberg-limited scaling*, *PRX Quantum* **6**, 040315 (2025). [doi:10.1103/j7b8-pb77](https://doi.org/10.1103/j7b8-pb77). Comparison uses [arXiv:2502.11900v2](https://arxiv.org/abs/2502.11900v2).
- [2] M. Shin, J. Lee, and C. Oh, *Heisenberg-limited Hamiltonian learning without short-time control*, [arXiv:2604.27838v1](https://arxiv.org/abs/2604.27838v1) (2026).
- [3] C. Cedillo Vayson de Pradenne, J. Cotler, and H.-Y. Huang, *Learning Hamiltonians at Long Times*, [arXiv:2606.05690v1](https://arxiv.org/abs/2606.05690v1) (2026).
- [4] T. Zhou and W. Gong, *Optimal Ansatz-free Hamiltonian Learning In Situ*, [arXiv:2606.19486v1](https://arxiv.org/abs/2606.19486v1) (2026).
- [5] M. Shin and Y. Tong, *Hamiltonian Learning and Certification via Eigenphase Engineering*, [arXiv:2609.26596v1](https://arxiv.org/abs/2609.26596v1) (2026).